

[Chathurya V R 251801045-2025-AIDS](#) ▾**C2****Started on** Thursday, 17 September 2026, 8:13 AM**State** Finished**Completed on** Thursday, 17 September 2026, 8:18 AM**Time taken** 4 mins 26 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main() {
4      int n;
5      // Read input from user
6      if (scanf("%d", &n) != 1) return 0;
7
8      int count = 0; // The step counter variable
9      int c = 0;    // The original algorithm's variable
10
11     // Step Counter Method Placement
12     count++; // 1 step for the initialization of the outermost loop (int i = n/2)
13
14     for (int i = n/2; (count++, i < n); i++) {
15         for (int j = 1; (count++, j < n); j = 2 * j) {
16             for (int k = 1; (count++, k < n); k = k * 2) {
17                 c++;
18                 count++; // 1 step for the execution of the body statement (c++)
19             }
20         }
21     }
22
23     // Print the total step count value
24     printf("%d\n", count);
25
26     return 0;
27 }
28
```

	Input	Expected	Got	
✓	4	30	30	✓

	Input	Expected	Got	
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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